

Internship - Software Engineer

Customs Integration Platform (CIP)

Role Purpose

As an intern in the Customs Integration Platform (CIP), you will gain hands-on experience building modern software solutions that support global customs, trade, and logistics processes. Working alongside experienced engineers, you will contribute to APIs, integrations, event-driven services, and cloud-native applications using technologies such as .NET, Kafka, Kubernetes, and modern cloud platforms.

We embrace an AI-first engineering culture where tools such as GitHub Copilot, Claude Code, and emerging developer productivity platforms are part of the day-to-day engineering experience. You will learn modern software engineering practices, experiment with cutting-edge technologies, and develop skills that are shaping the future of software development while delivering real business value.

Why Join Us?

- Build software supporting global customs and trade operations at scale.
- Gain hands-on experience with .NET 10, Kafka, Kubernetes, microservices, and cloud-native architectures.
- Learn how AI-powered engineering tools are transforming software development and developer productivity.
- Work with experienced engineers, receive mentorship, and contribute meaningful code from day one.

Key Responsibilities

- Contribute to the software development lifecycle across discovery, design, development, testing, deployment, operations, and continuous improvement of CIP capabilities.
- Learn and apply modern software engineering practices while building real business solutions with guidance from experienced engineers.
- Design, build, test, and operate scalable APIs, integrations, event-driven services, and cloud-native platform capabilities.
- Participate in design discussions and code reviews with focus on security, maintainability, scalability, reliability, and user experience.
- Contribute to high-quality delivery through automated testing, CI/CD, peer reviews, and adherence to engineering standards.
- Collaborate with engineers, product teams, and business stakeholders across global and cross-functional teams.
- Support releases, cutovers, operational activities, and troubleshooting with appropriate guidance.
- Share learnings, contribute improvement ideas, and grow through mentorship, retrospectives, and engineering communities.

Technical Capabilities

- Foundation in software engineering principles, object-oriented programming, algorithms, data structures, and design patterns.

- Experience with C# and modern .NET development (.NET 10 and beyond) through academic, personal, or internship projects.
- Understanding of web APIs, RESTful services, microservices, and modern software architecture concepts.
- Familiarity with cloud-native development, containerization technologies such as Docker and Kubernetes, and scalable application design.
- Exposure to relational and NoSQL databases, data modeling, and data access patterns such as SQL, document databases, and Cosmos DB.
- Understanding of event-driven architecture, asynchronous communication, and messaging or streaming technologies such as Apache Kafka.
- Familiarity with source control, automated testing, CI/CD concepts, Agile, TDD, and BDD.
- Interest in software quality, observability, performance, reliability, security, operational excellence, and AI-assisted development tools.

AI-Assisted Engineering Expectations

- Use AI-powered software engineering tools such as GitHub Copilot, Claude Code, and emerging developer productivity platforms to support development, testing, debugging, documentation, and learning.
- Apply an AI-first engineering mindset to accelerate delivery, improve code quality, automate repetitive tasks, and increase development effectiveness.
- Explore new AI capabilities responsibly, share learnings with peers, and help shape modern engineering practices across the platform.
- Use AI tools in a secure and responsible way while maintaining high standards of quality, reliability, and engineering discipline.

Ideal Candidate Profile

- Currently enrolled in a Bachelor's degree program in Computer Science, Software Engineering, or a related technical field; ideally in the later stage of studies, such as 4th year.
- Strong programming fundamentals and enthusiasm for software engineering, distributed systems, cloud-native applications, APIs, and event-driven architecture.
- Curious, self-motivated, and comfortable learning new technologies, tools, and engineering practices.
- Strong problem-solving, troubleshooting, analytical thinking, and attention to detail.
- Good written and verbal communication skills, with ability to collaborate in diverse, global, and cross-functional teams.
- Takes ownership of learning, seeks feedback, and continuously improves through hands-on experience.
- Open to experimentation, innovation, and responsible use of AI-assisted software engineering tools.
- Interested in contributing to modern customs technology solutions that support global trade and logistics.